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Revision History		
REV	REVISION NOTES	Date
X1	Initial	Jun 10, 2023
A	Final Release	Jul 10, 2023
B	1.Change 3pin solder pads. 2.Update schematic per Ver.A test result.	Sep 15, 2023

FRDM-MCXN947

<Core Design>



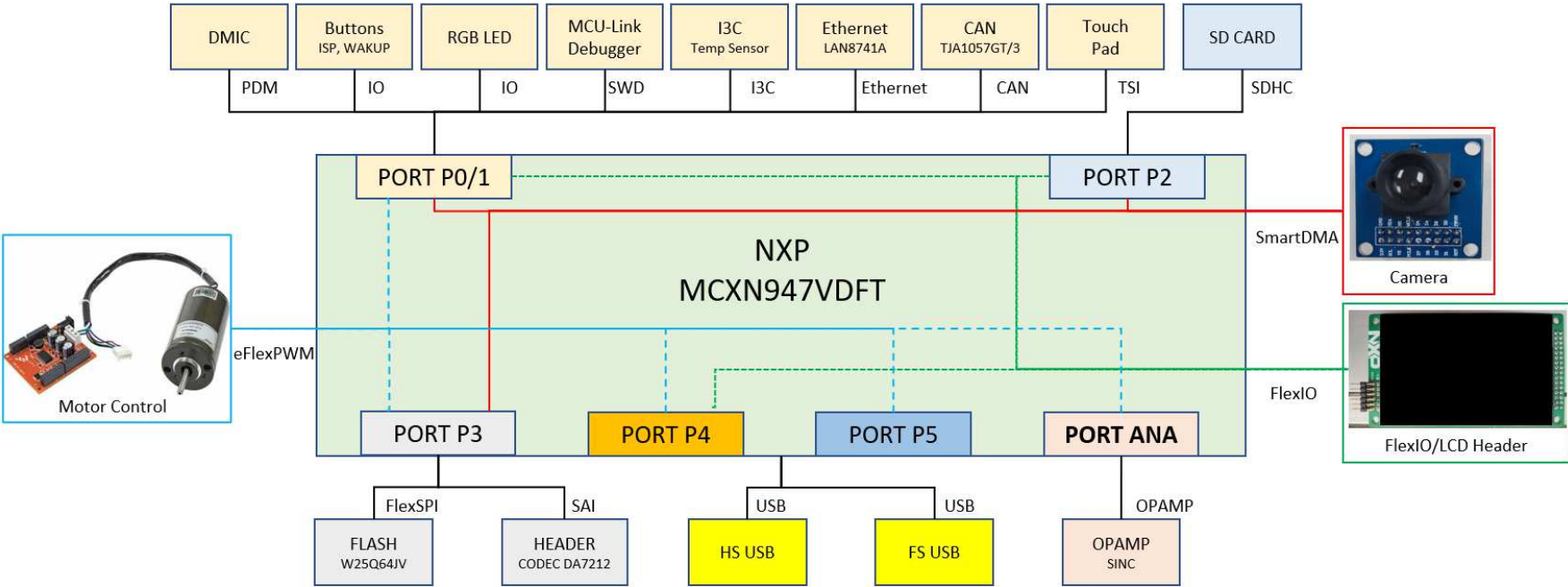
Microcontroller Product Group
6501 William Cannon Drive West
Austin, TX 78735-8598

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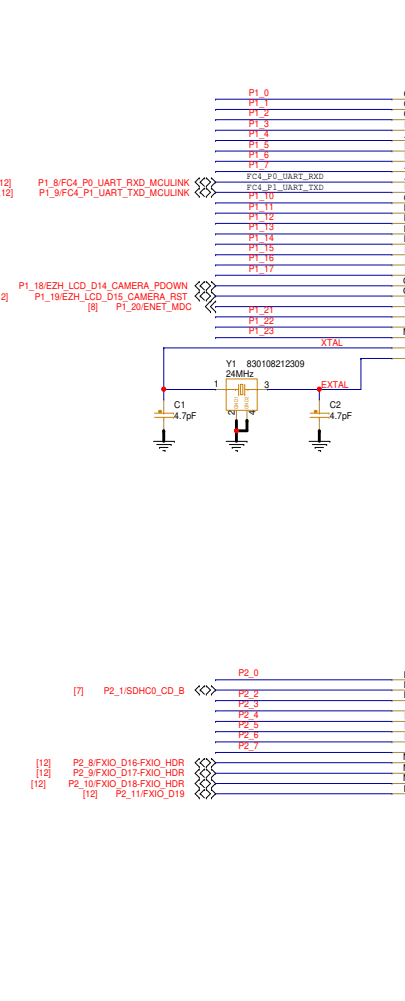
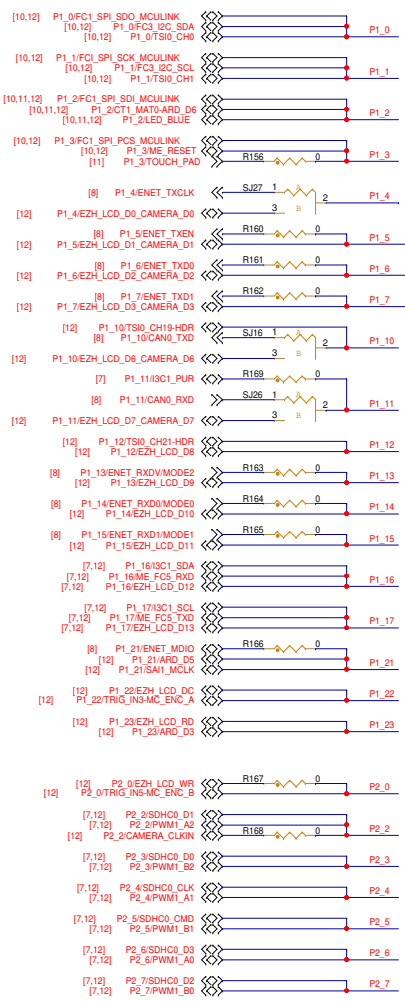
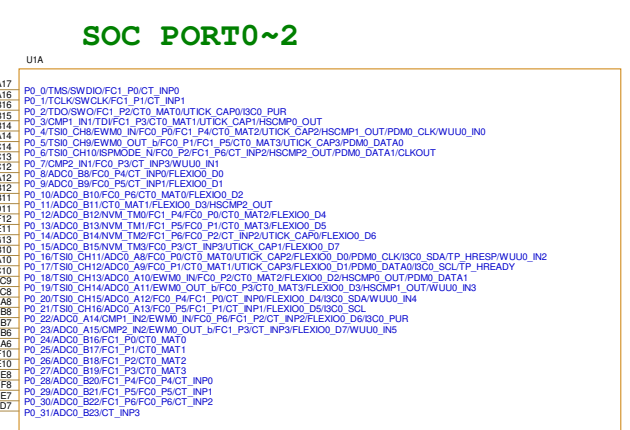
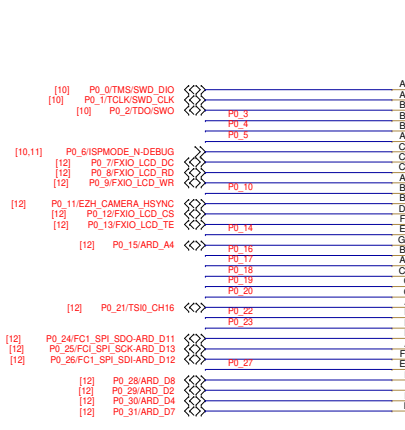
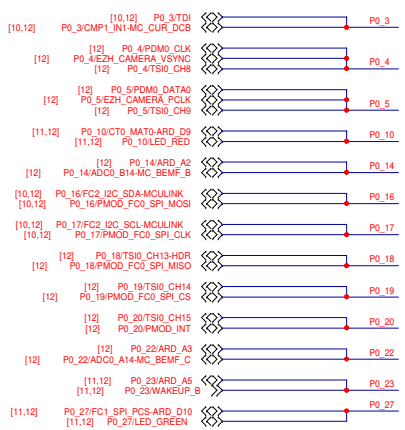
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Designer: John Wu	Drawing Title: FRDM-MCXN947		
Drawn by: Kate Fan	Page Title: TITLE PAGE		
Approved: William Jiang	Size C	Document Number SCH-90618 PDF: SPF-90618	Rev B
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BLOCK DIAGRAM



SOC PORT0~2



SOC PORT3~5



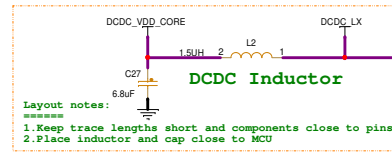
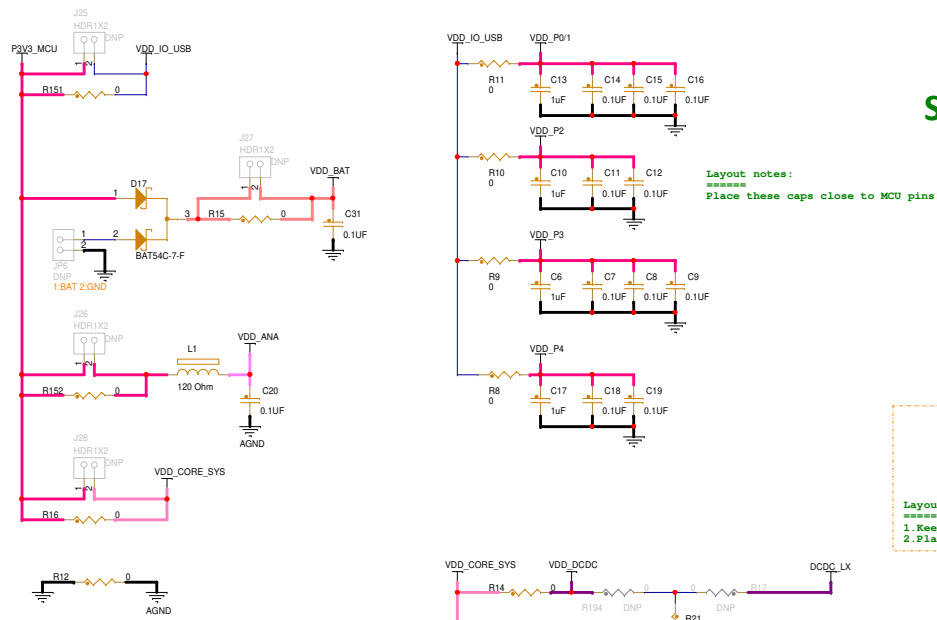
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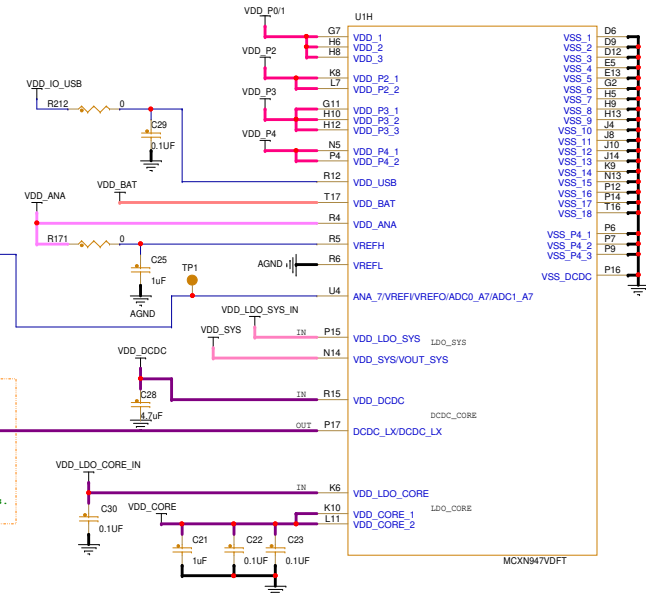
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SOC POWER

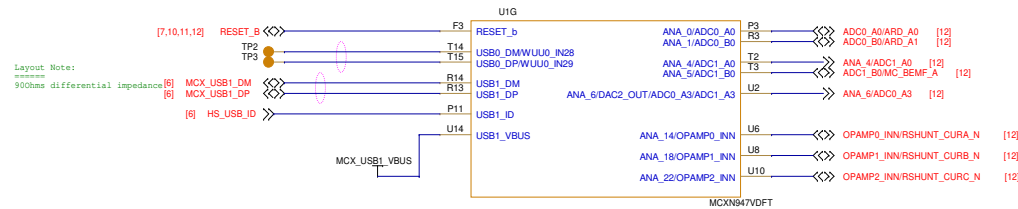


Mode	Solder Option
DCDC_CORE---Enable LDO_CORE---Disable (Default)	Populate R14, R18, R19; DNP R17, R194, R20.
DCDC_CORE---Disable LDO_CORE---Enable	Populate R17, R194, R20; DNP R14, R18, R19.

DNP R23(default): Enable internal LDO_SYS.
 Solder R23: Disable internal LDO_SYS.



SOC USB & Analog Signals



<Core Design>



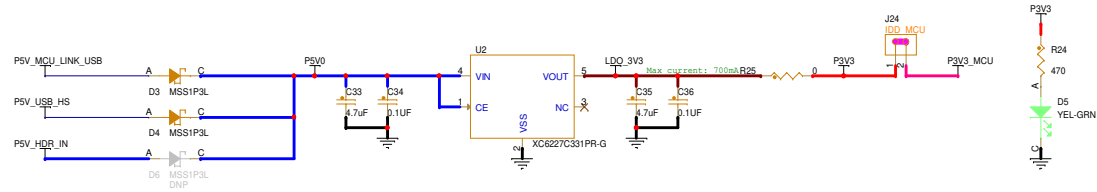
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Drawing Title: **FRDM-MCXN947**

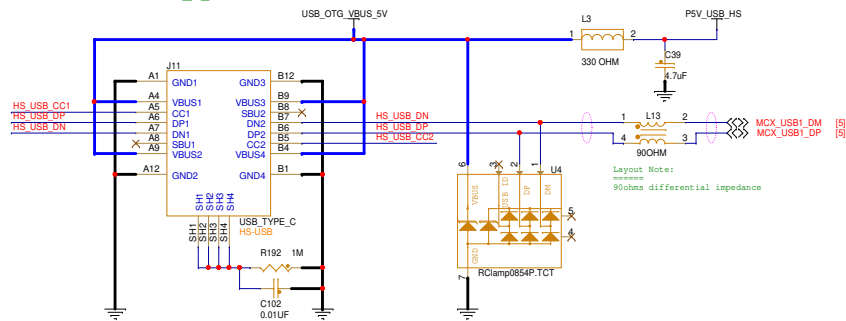
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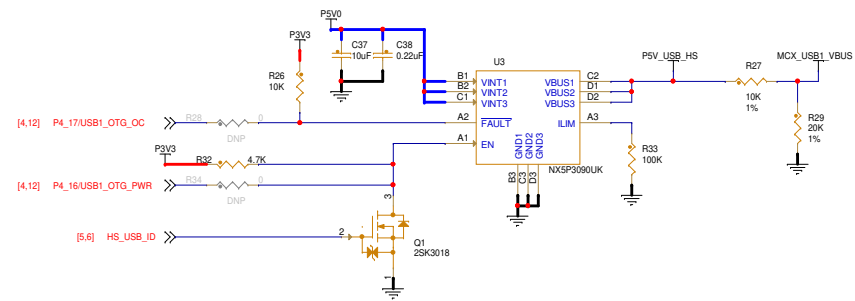
SYSTEM POWER



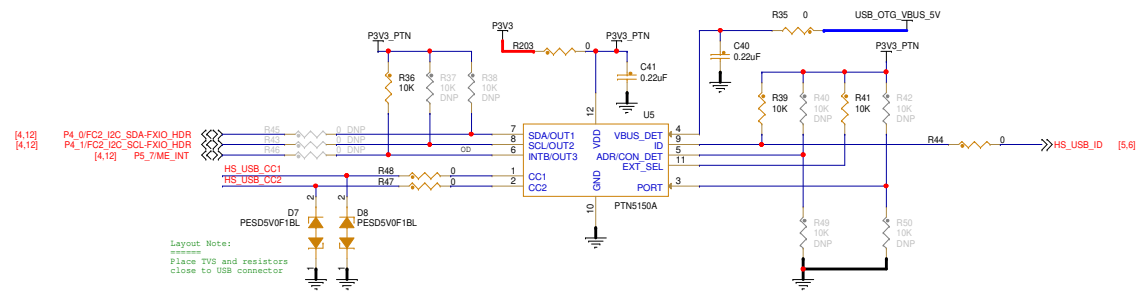
USB1_HIGH SPEED
USB2.0 Type C



VBUS Power Control



CC Logic



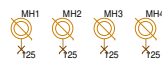
ADR/CON_DET:
When Power-up, ADR(input) function:
ADR=1: I2C Address: 0x7A(ADR)
ADR=0: I2C Address: 0x3A(ADR)
ADR=MID/FLOATING
After TINPUTLATCH, CON_DET(output) function:
CON_DET=1: Connection Detected
CON_DET=0: No Connection

EXT_SEL: External selection
High = CC1 orientation or no valid CC1/CC2 detection
Low = CC2 orientation

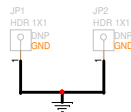
PORT=:

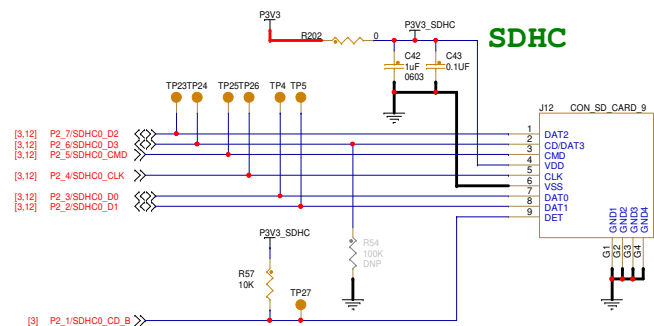
- 1: DFP mode
- 0: UFP mode
- Floating: DRP mode

MOUNTING HOLE



Test Points

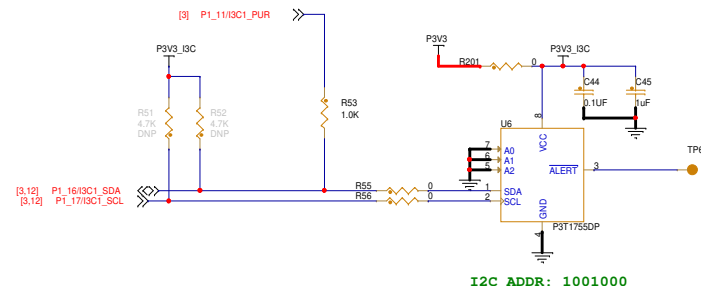




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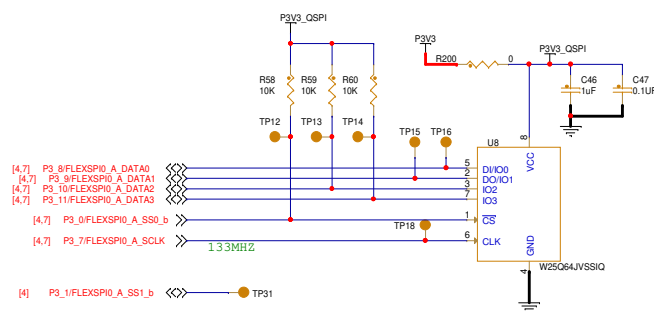
- 1) SDHC signals: equal length routing with 50ohm resistance. And as short as possible.
- 2) Place R54 to easily rework.

I3C SENSOR



I2C ADDR: 1001000

QSPI



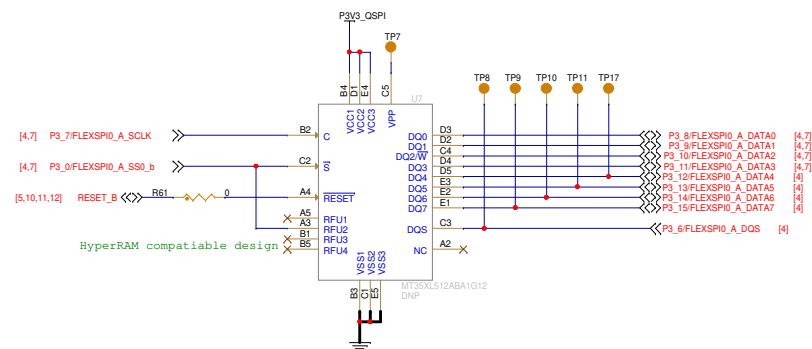
Other QSPI Flash Option: MT25QL128ABA1ESE-0SIT (MICRON)

Layout Notes:

=====

- 1.U7 & U8 footprint overlapped.
- 2.Equal length routing with 50ohm resistance.
- 3.Place TP31 close to TP12.

Octal Flash



<Core Design>



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Drawing Title: **FRDM-MCXN947**

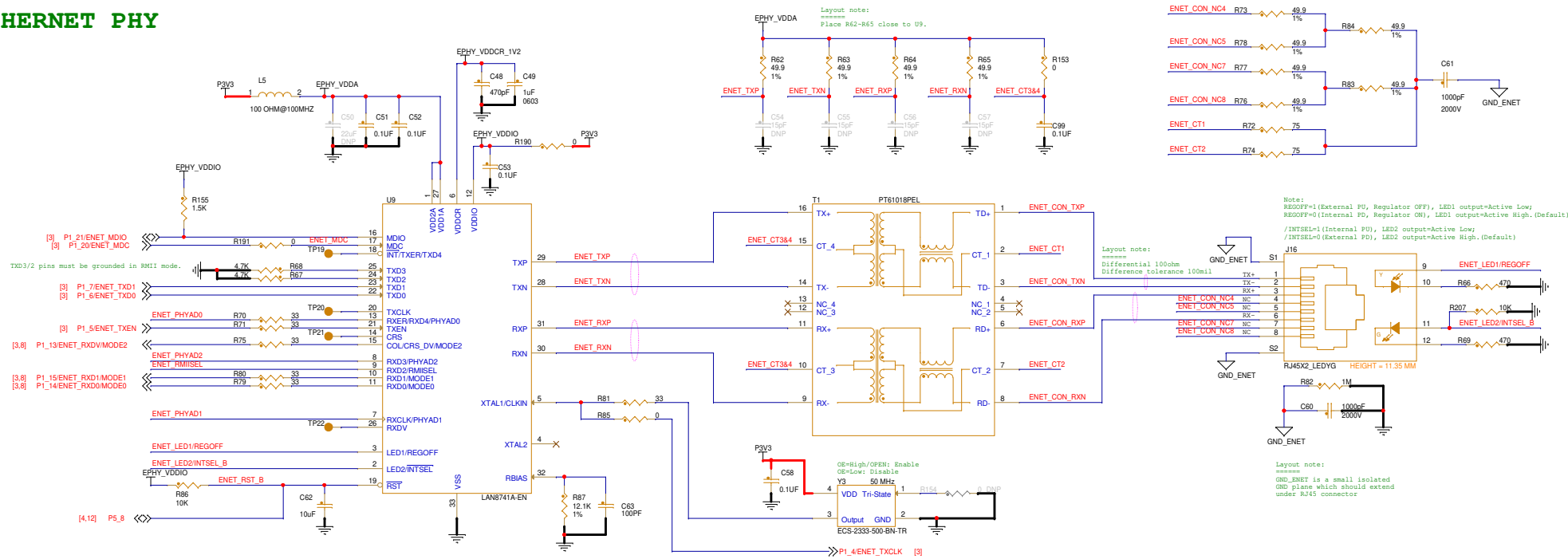
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Size C Document Number SCH-90818 PDF: SPF-90818

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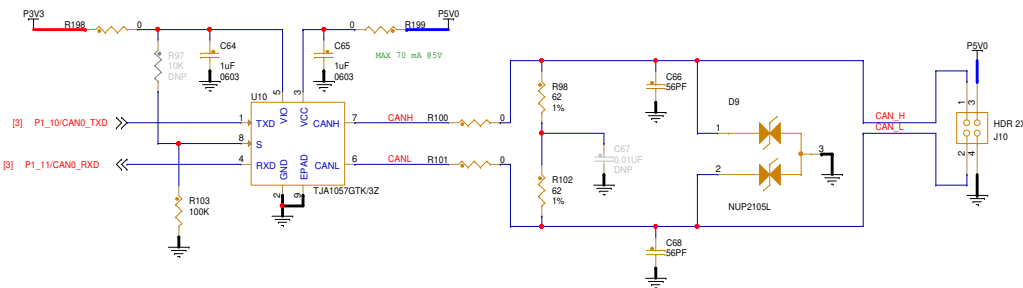
Rev B

ETHERNET PHY



Config Pin (Internal State)	Description	Jumper Setting
MODE[0:2] (PU)	Mode Config 111 => All Capable, auto-neg enabled (Default)	Open (Int PU)
PHYAD[0:2] (PD)	Device Address 000 => Default	Open (Int PU)
RMIISEL (PD)	1 => RMII (Default) 0=> MII	Closed (Ext PU)
REGOFF (PD)	1 => Internal Regulator OFF 0/Floating=> Internal Regulator ON (Default)	Open (Int PU)
INTSEL_B (PU)	Floating 1 => nINT function (Default) 0=> TXD4/TXER function	Open (Int PU)

CAN PHY



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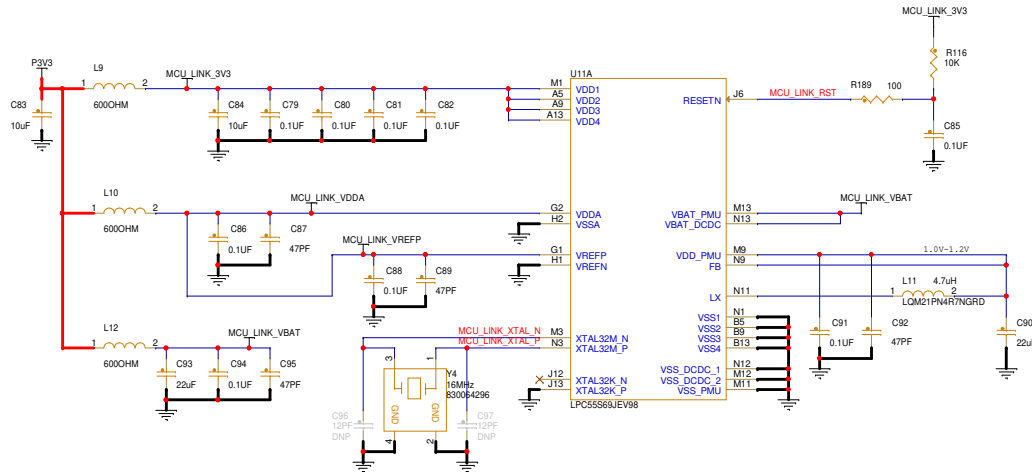
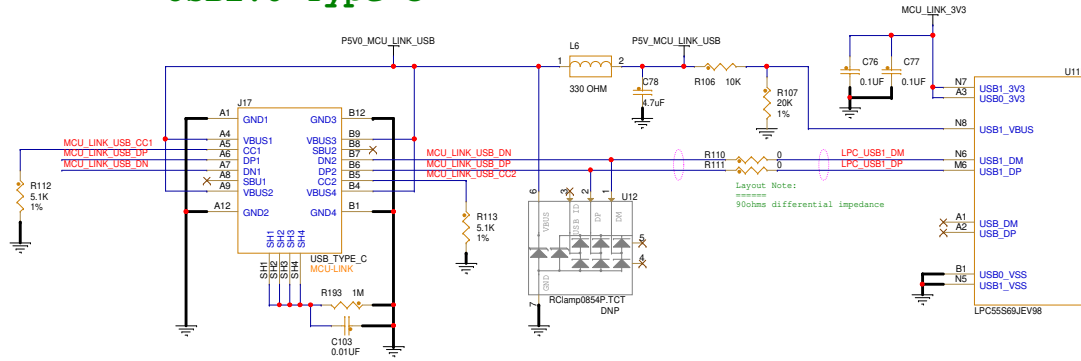
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Size C	Document Number SCH-90818 PDF: SPF-90818
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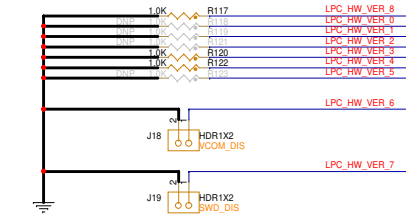
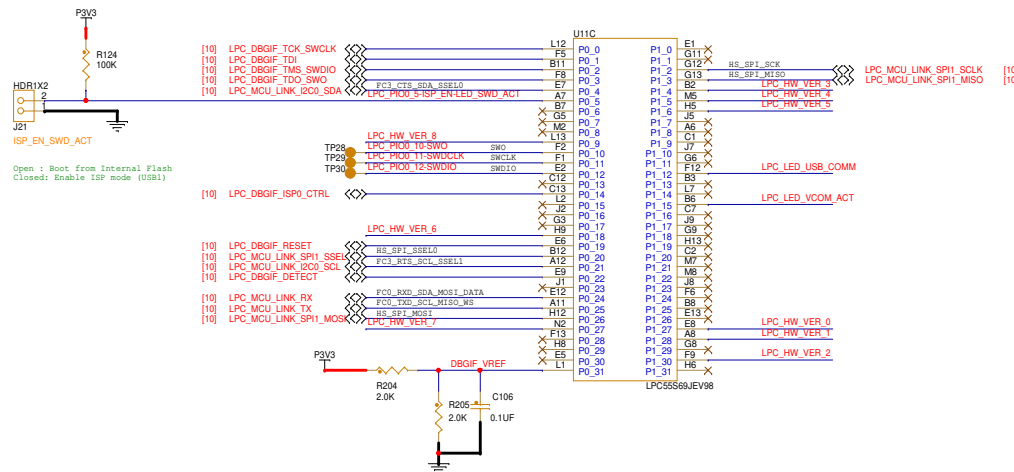
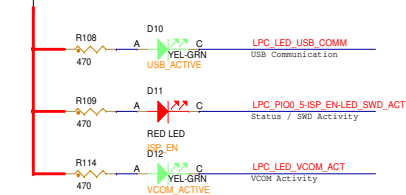
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MCU-Link USB2.0 Type C

Layout Note:
Add a outline for MCU LINK related components.



MCU-Link LEDs



HW_VER_0: USB power negotiation enable when low
HW_VER_1: USB-SIO bridge disable when low
HW_VER_2: OS selection when low
HW_VER_3: Board identity code used when low
HW_VER_4: Power measurement disable if leave open
HW_VER_5: VCOM disable when low
HW_VER_6: SMD disable when low
HW_VER_7: Secondary VCOM disable when low



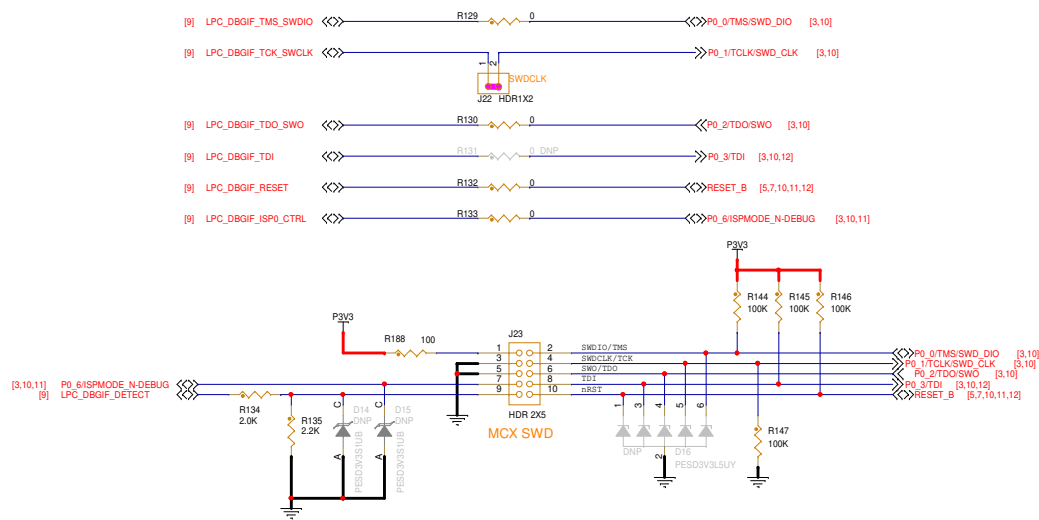
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FRDM-MCXN947

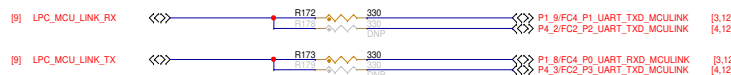
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MCU_LINK_USB

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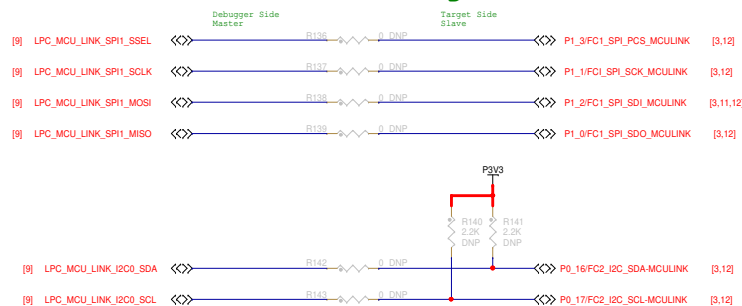
MCU-Link Debug Interface



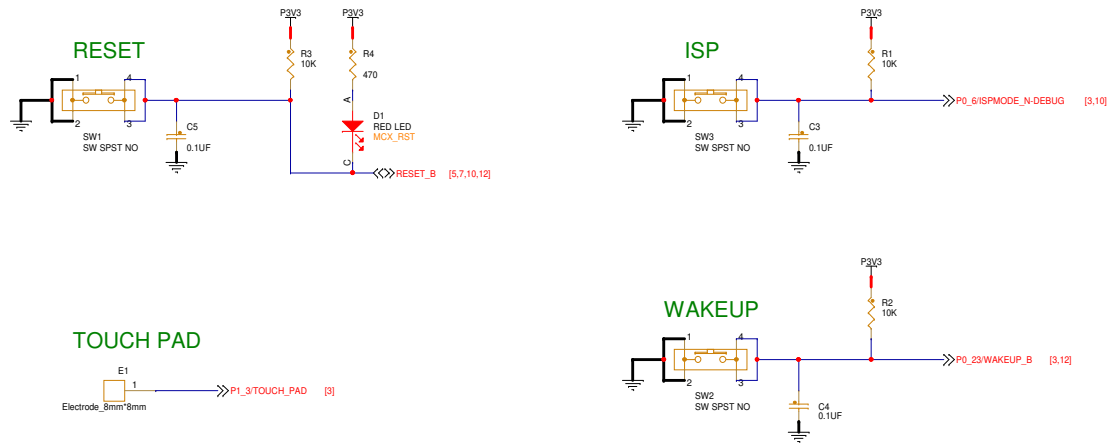
MCU-Link UART



USB Bridge



HMI



RGB



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Drawing Title:
FRDM-MCXN947

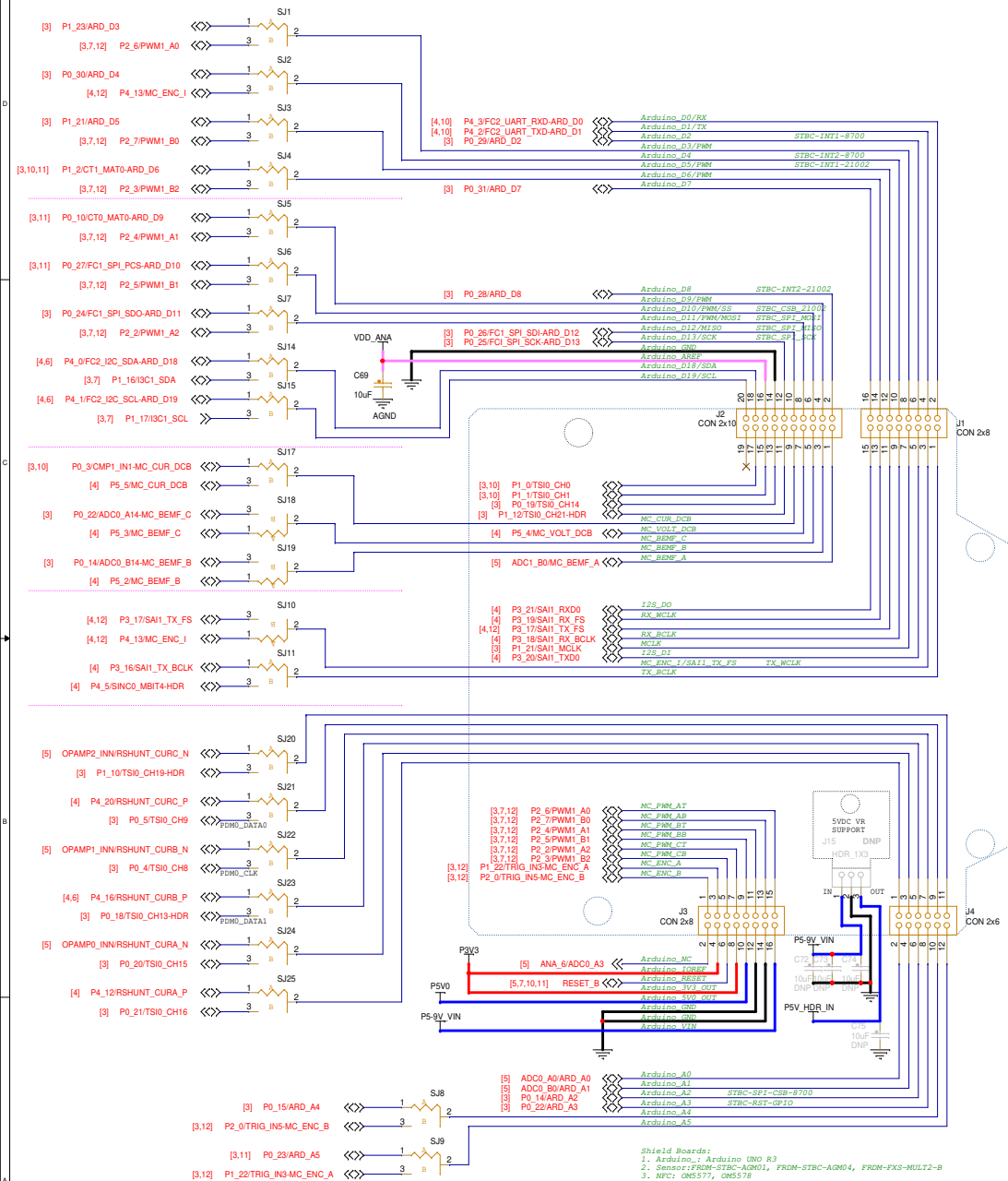
Page Title:
BLOCK DIAGRAM

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ARDUINO SHIELD COMPATIBLE HEADERS



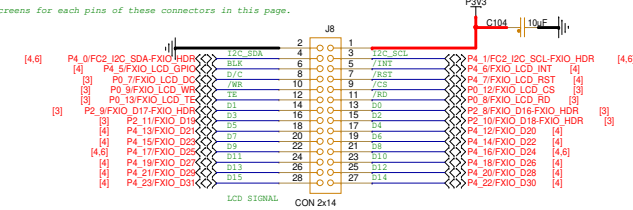
Shield Boards:

1. Arduino: Arduino UNO R3
2. Sensor: FRDM-STBC-AGM01, FRDM-STBC-AGM04, FRDM-FXS-MUL72-B
3. NFC: CM5577, CM5578
4. WiFi: Mikro Click connections
5. Motor control: FRDM-MC-LVBLDC, FRDM-MC-LVPM3M, Arduino-SimpleFOCShield
6. Touch: FRDM-TOUCH
7. Audio: ARD-AUDIO-DA7212

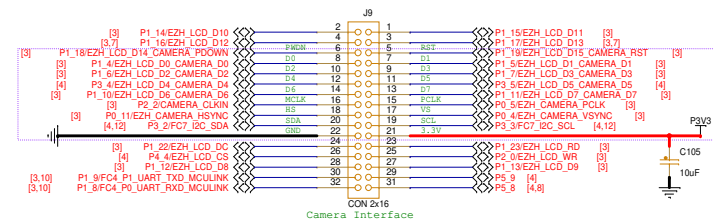
FlexIO LCD

Layout notes:

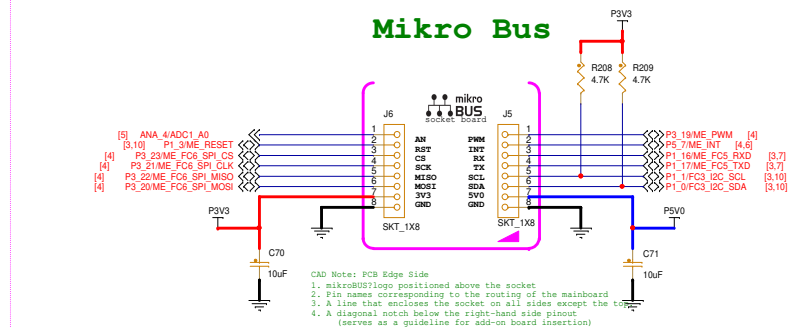
Add clear silkscreens for each pins of these connectors in this page.



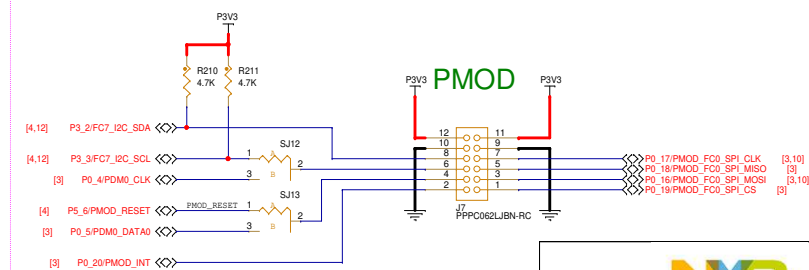
Camera



Mikro Bus



P3V3 PMOD



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Drawing Title: **FRDM-MCXN947**

Page Title:	HEADERS
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REF DES	JUMPER(DEFAULT)	PAGE NAME
J24	1-2	06 USB & PWR
J18,J19	OPEN	09 MCU_LINK_USB
J22	1-2	10 MCU_LINK_DEBUG

REF DES	ASSY_OPT	PAGE NAME
J25,J26,J27,J28,JP6,R17,R20,R23,R194	DNP	05 SOC_PWR
D6,JP1,JP2,R28,R34,R37,R38,R40,R42,R43,R45,R46,R49,R50	DNP	06 USB & PWR
J12,R51,R52,R54,U7	DNP	07 SD & QSPI & SENSOR
C50,C54,C55,C56,C57,C67,R88,R89,R90,R91,R92,R93,R95,R96,R97,R154	DNP	08 CAN PHY & ETHERNET PHY
C96,C97,R118,R119,R121,R123,U12	DNP	09 MCU_LINK_USB
D14,D15,D16,R131,R136,R137,R138,R139,R140,R141,R142,R143,R178,R179	DNP	10 MCU_LINK_DEBUG
C72,C73,C74,C75,J7,J15	DNP	12 HEADERS

REF DES	SHORT(DEFAULT)	PAGE NAME
SJ16,SJ26,SJ27	1-2	03 SOC_PORT0~2
SJ11,SJ2,SJ3,SJ4,SJ5,SJ6,SJ7,SJ10,SJ11,SJ12,SJ13,SJ14,SJ15,SJ17,SJ18,SJ19,SJ20,SJ21,SJ22,SJ23,SJ24,SJ25	1-2	12 HEADERS

APPENDIX JUMPER/DNP

